

Rachel Kan

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Education

Cornell University, College of Engineering, GPA: 3.87

Ithaca, NY

Bachelor of Science in Computer Science • Intended Minors: Artificial Intelligence, Business

Expected May 2028

Related Courses: Honors Object-Oriented Design and Data Structures, Data Structures and Functional Programming, Discrete Structures, Probability and Statistics, Foundations of AI Reasoning and Decision-Making, Intro to Machine Learning, Intro to Analysis of Algorithms, Language and Information, Intro to Database Systems (In-Progress)

CS 4820/5820: Introduction to Analysis of Algorithms Teaching Assistant

January-May 2026

CS 3700/5700: Foundations of AI Reasoning and Decision-Making Teaching Assistant

August 2026-Present

Technical Skills

- **Programming:** Python, Java, C++, C, OCaml, JavaScript, HTML/CSS, QML, JSON, RESTful APIs
- **Frameworks/Tools:** Git, PyTorch, Pandas, NumPy, NLTK, Sentence-Transformers, NetworkX, Jupyter, React, AI-assisted development

Project Experience

Knowledge Graphs for Solar Energy News Analysis

June-August 2026

Research Assistant, Syracuse University Text Analytics for Social Scientists (NSF/DOD-funded)

Syracuse, NY

- Built a Python pipeline converting unstructured solar news into a 4,002-node GPT-4o knowledge graph
- Cut LLM inference costs via domain-specific keyword filtering and preprocessing (Pandas, NLTK, NumPy)
- Resolved entities via Sentence-Transformers embeddings and RapidFuzz matching, graphed with NetworkX
- Applied Clauset-Newman-Moore detection to surface 408 communities; presented at a research poster session

FloraSense App: Information Retrieval with RAG

January-May 2026

CS 4300: Language and Information Final Project

Ithaca, NY

- Augmented scientific data sources with web scraping to create a flower dataset with flower language and meaning
- Implemented SVD and TF-IDF similarity measurements to compare user queries with available flower data
- Integrated LLM with retrieval methods to summarize flower results and explain matches

AI Sector Financial Analysis and Sentiment Modeling for AI Bubble prediction

August 2025-May 2026

Quantitative Student Analyst, Millennium x WICC Project

Ithaca, NY

- Collaborated in a 4-person team under the guidance of a Millennium Technology Project Lead to conduct analysis of trends in AI sector stocks and company valuations and presented findings at Millennium headquarters
- Applied RoBERTa and VADER to news, finance, and social streams to correlate sentiment with market moves
- Indexed AI stock trajectories against the Dot-Com crash (yfinance), synthesizing capex and multiples

Real-Time Command and Control Graphical User Interface (GUI)

June 2025-August 2025

Software Engineering Intern, Advanced Technology Systems Company (ATSC-IES)

Gilbert, AZ

- Designed and implemented C++ RESTful APIs exposing real-time PTZ camera control (JSON-over-RTSP) and server-side spatial analysis, consumed by both touchscreen and desktop clients
- Built the map rendering path for streaming MIL-STD-2525 data, clustering tracks to hold frame rates
- Implemented viewshed, tape measure, and pin drop geometry with direct drag-edit of placed shapes
- Architected a Qt client syncing a live RTSP video feed against real-time map views

Extracurricular Activities

Women in Computing at Cornell (WICC) Executive Board Member

January 2025-Present

- Underclassmen Outreach Co-Director
- Vice President: Academic Team

August 2025-May 2026

May 2026-Present

Skills and Interests

Language Skills: Conversational Fluency in Mandarin, Professional Fluency in Spanish

Interests: Nonograms, Nail art, JoJo's Bizarre Adventure